ProgressiveScorer: 每 Agent · 6维 EWMA/CUSUM 渐进跟踪 → force_observer

L1 规则引擎
Rule Engine · YAML 规则集热加载

L2 图模式匹配
Graph Pattern Match · 子图同构

L3 加权评分
Weighted Scoring · 多维融合

6-Dim Fusion
7子分数 → OWA(80%) + GeoMean(20%)

Platt 概率校准
P = 1/(1+e^{-(7.0×S−3.0)})

Evidence Gate
PBC < 10 且无 Unsafe → cap 0.10

Safe
S_cal < 0.15

S_cal ∈ [0, 1]

Direct Unsafe
S_cal > 0.70

Grey Zone: 0.15 ≤ S_cal ≤ 0.70
模糊区间 → 根据 force_observer 标志选择路由

force_observer = true

force_observer = false

Observer LLM 语义仲裁 (冷路径)
Unsafe → S×0.4 + conf×0.6, Safe → S×(1 − conf×0.7)

Phase 2 启发式回退 (批量路径)
fail-closed: Grey Zone → Unsafe / Warning

ClassificationResult → Claims → Alerts